

Algorithmic Trading Strategy using Relative Strength Index (RSI)

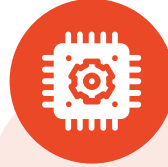
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Relative Strength Index (RSI)



RSI

- Measures a stock's momentum, indicating speed and size of price changes.
- Helps identify if a stock is overbought or oversold.
- Aids in determining potential entry and exit trading signals.



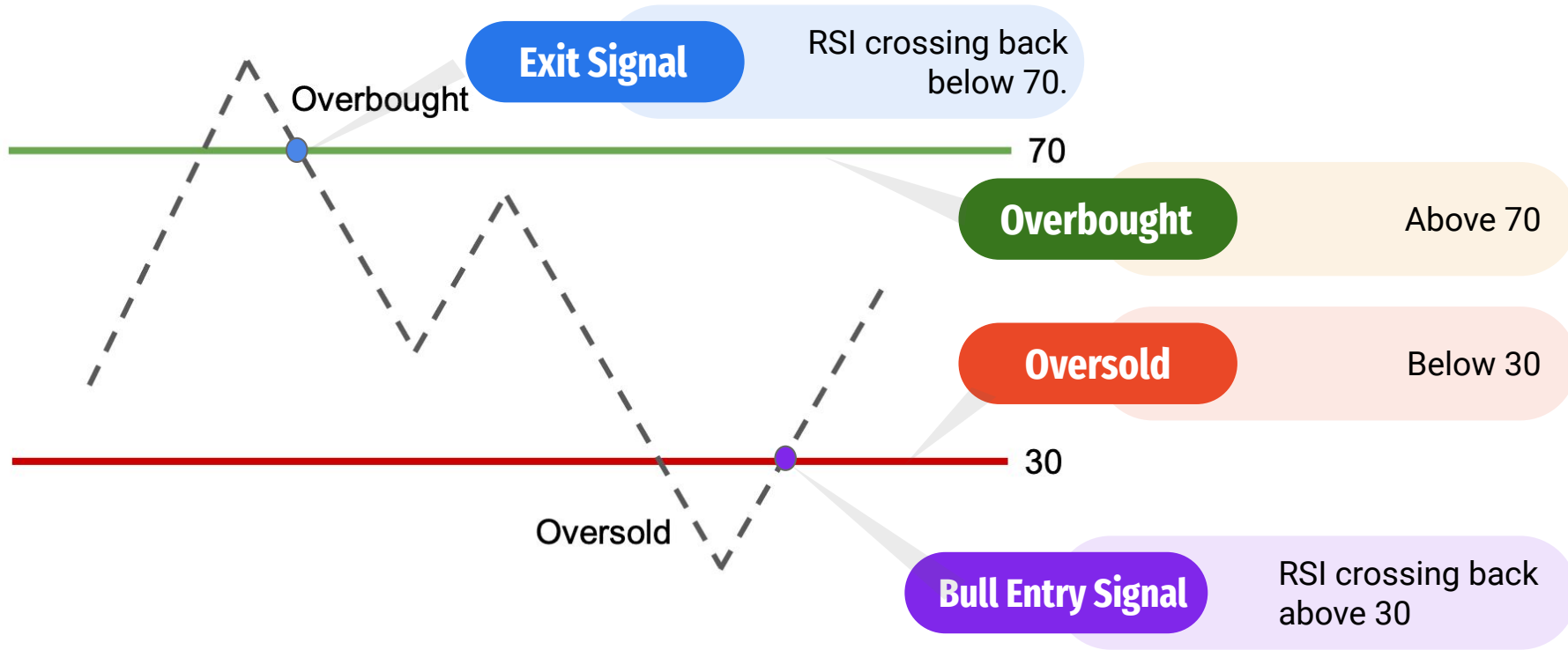
How to Calculate

$$RSI = \frac{Avg.gain}{Avg.loss}$$

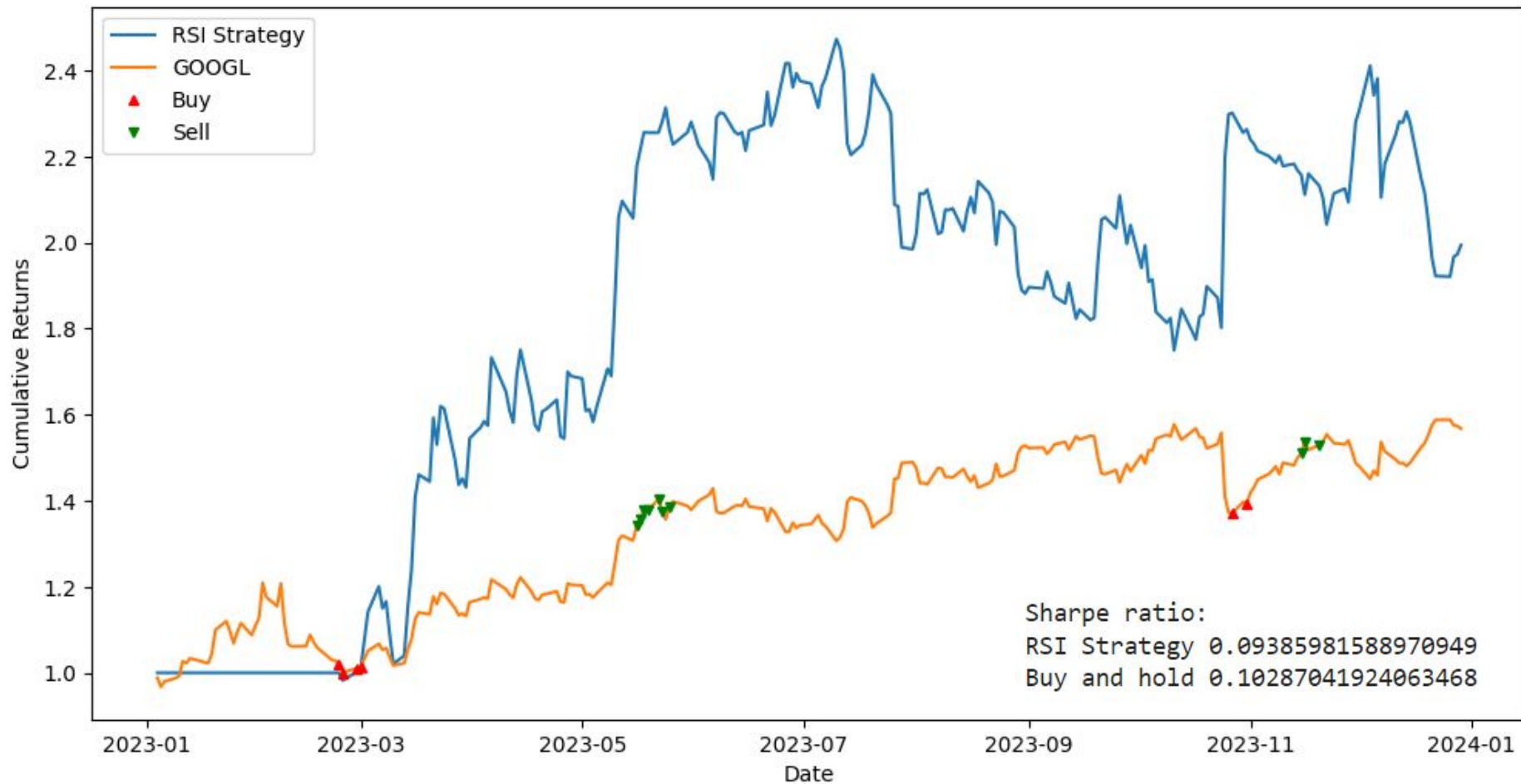
- Define a time period (typically 14 days).
- Calculate average gains and average losses.
- Plot relative strength value on a graph between 0 and 100.

Using RSI for Trading Signals

The RSI

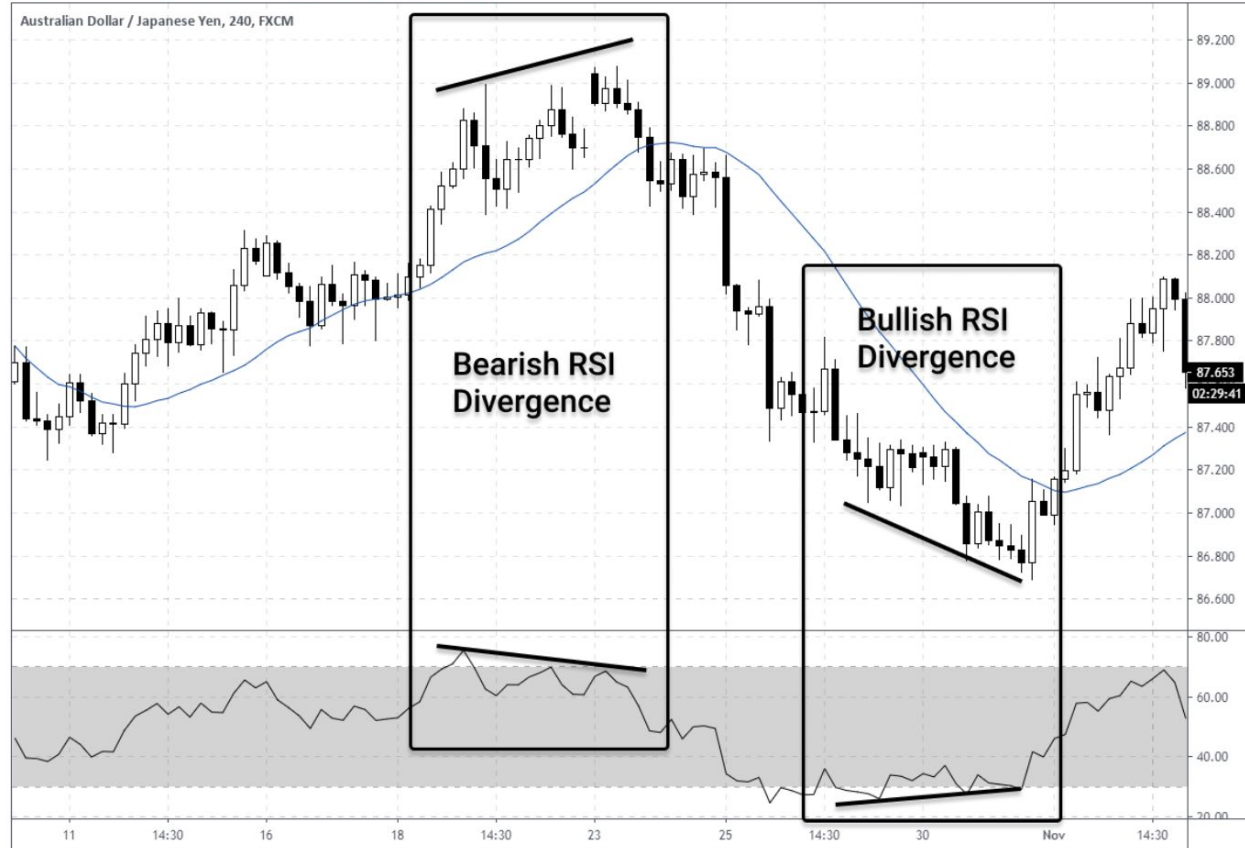


Cumulative Returns of RSI vs. Buy-and-hold GOOGL stock in 2023.



Spotting Trend Reversals

RevenueTrader published on TradingView.com, April 29, 2020 20:00:22 IST
FX:AUDJPY, 240 69.599 ▲ +0.235 (+0.34%) O:69.595 H:69.707 L:69.520 C:69.599



Divergence: Opposite movement between stock price and RSI.

Bullish Divergence: Lower lows in stock price, higher lows in RSI.

Bearish Divergence: Higher highs in stock price, lower highs in RSI.

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RSI Limitation

Imperfect

There are always external factors.

01

Prolonged Overbought/Oversold

Inactive trading period

02

Improvement

With other indicators for better decision-making

03

